

Machine Learning Workflows for Digital Energy Systems
(SET3125) 2024/2025 Q2
Project Electrical System

Group 2
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Basic Workflow with Fully Connected Neural Network (FNN)

Task 1

Check the distributions of the load demands and generator supplies, are there outliers? Comment on how you would remove them if any. Outliers can be best seen in the load demand distributions of individual buses. As per example, by looking at the histograms of the first three buses in Figure 1, the outer boxes on both sides can be regarded as outliers, deviating markedly from the overall pattern of the rest of the data. The removal of these outliers has been

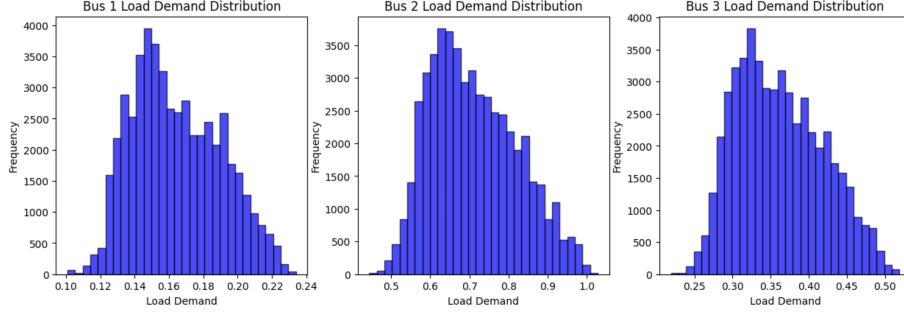


Figure 1: Load demand distribution of the first three busses.

done using the interquartile range (IQR) method because of its simplicity and broad applicability on datasets. IQR identifies the spread of the middle 50% of the data. Depending on this spread and a K factor, outliers at both ends are removed. $K = 3$ was chosen to remove only the extreme outliers in the dataset. It is commonly used for this purpose as it strikes a balance between detecting outliers without being overly sensitive. The same approach has been used for the generator data. **Look at the total demand for each data sample. Does it match total supply?** Observing the plot in Figure 2a, it's evident that they generally nearly match, but there are some bigger mismatches, especially on the shortage side.

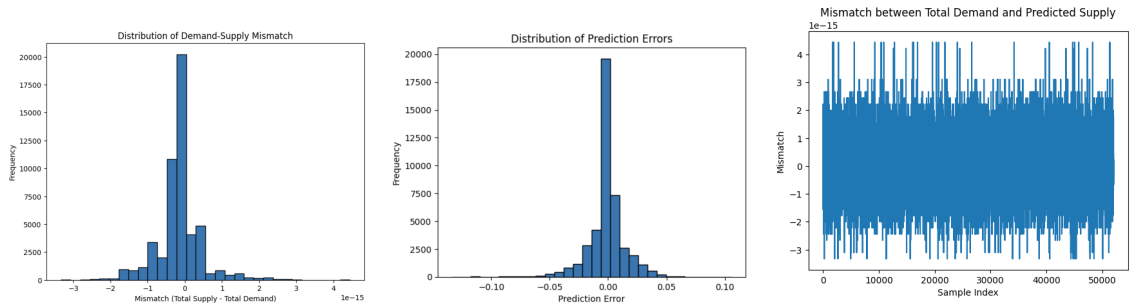
Task 5

Comment on the influence of model hyperparameters on the model accuracy. Hyperparameters can drastically influence a model's accuracy by affecting everything from how well the model fits the data (bias-variance tradeoff) to how quickly or stably it converges during training. The two hyper-parameters controlled in the FNN are the *number of hidden features* and *layers*. Increasing the number of hidden features from 32 to 64 generally improved the model's accuracy, indicating that the model benefits from more capacity. In fact, the best performance was achieved with 64 hidden features and 3 layers, suggesting a good balance between model complexity and generalization.

Task 6

Plot a histogram of prediction errors, what type of distribution do you obtain? The resulting plot in Figure 2b shows a normal distribution.

For the test data, plot the mismatch between the sum of demands and the sum of predicted generator supply. What is the maximum mismatch? What is the mean mismatch? The resulting plot is shown in Figure 2c. The maximum power balance mismatch is: 0.165. The mean power balance mismatch is: 0.003



(a) Demand-Supply mismatches across samples.

(b) Prediction errors histogram

(c) Mismatch between sum of demands and sum of predicted generator supply

Figure 2

Physics-Informed Training

Task 1

Write the physics-informed loss function using the power balance mismatch function provided. The function can be observed in Figure 3.

```
[ ] def extra_loss(x, y_pred): #Calculates the physics-informed loss based on total power balance (No data about topology).  
  
    total_demand = torch.sum(x, dim=1) #Sum demand across all buses  
    total_supply = torch.sum(y_pred, dim=1) #Sum supply across all generators  
  
    residual = total_supply - total_demand #Total power mismatch  
    phy_loss = torch.mean(residual**2) #Mean squared error of the mismatch  
  
    return phy_loss * physics_loss_weight #Scale by physics loss weight
```

Figure 3: Physics informed training python function

Task 4

Check how well the power balance constraint is satisfied now with the physics loss included by computing the maximum and mean power balance mismatches. The maximum power balance mismatch is: 0.064. The mean power balance mismatch is: 1.706×10^{-5} . Plot the error distribution and comment on the type of distribution. The error distribution for a fully connected neural network (Figure 4a) resembles a normal distribution.

GNN Training

Task 5

Finally for the GNN model, check how well the power balance constraint is satisfied by computing the maximum and mean power balance mismatches. The maximum power balance mismatch is: 36.932. The mean power balance mismatch is: 29.723.

Plot the error distribution and comment on the type of distribution. As for the other models, the error distribution (Figure 4b) represents a normal distribution.

Task 6

Which of the two models is more parameter efficient? GNN Model Parameters: 122565 Physics-informed Training Model Parameters: 9605 The Physics-informed training model is more parameter-efficient.

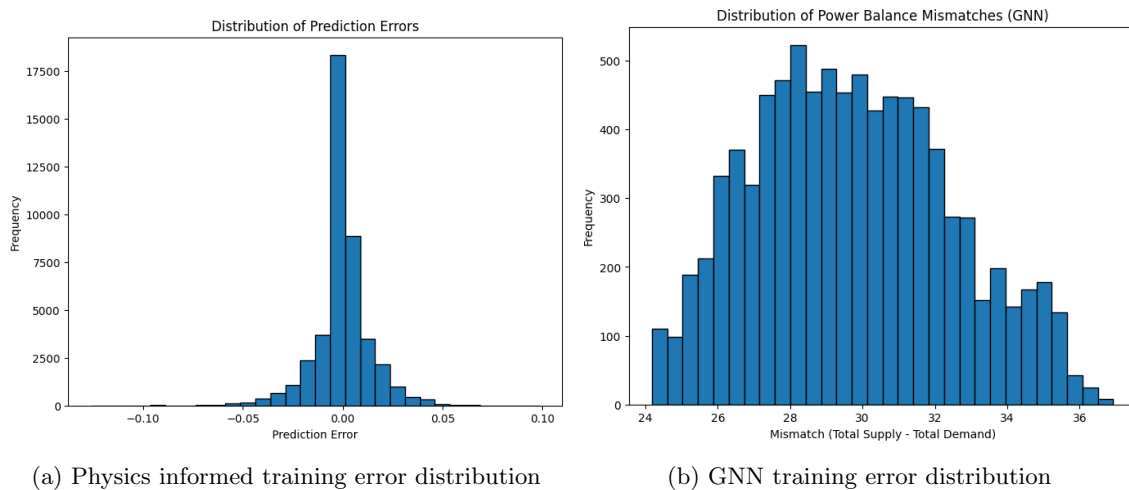


Figure 4